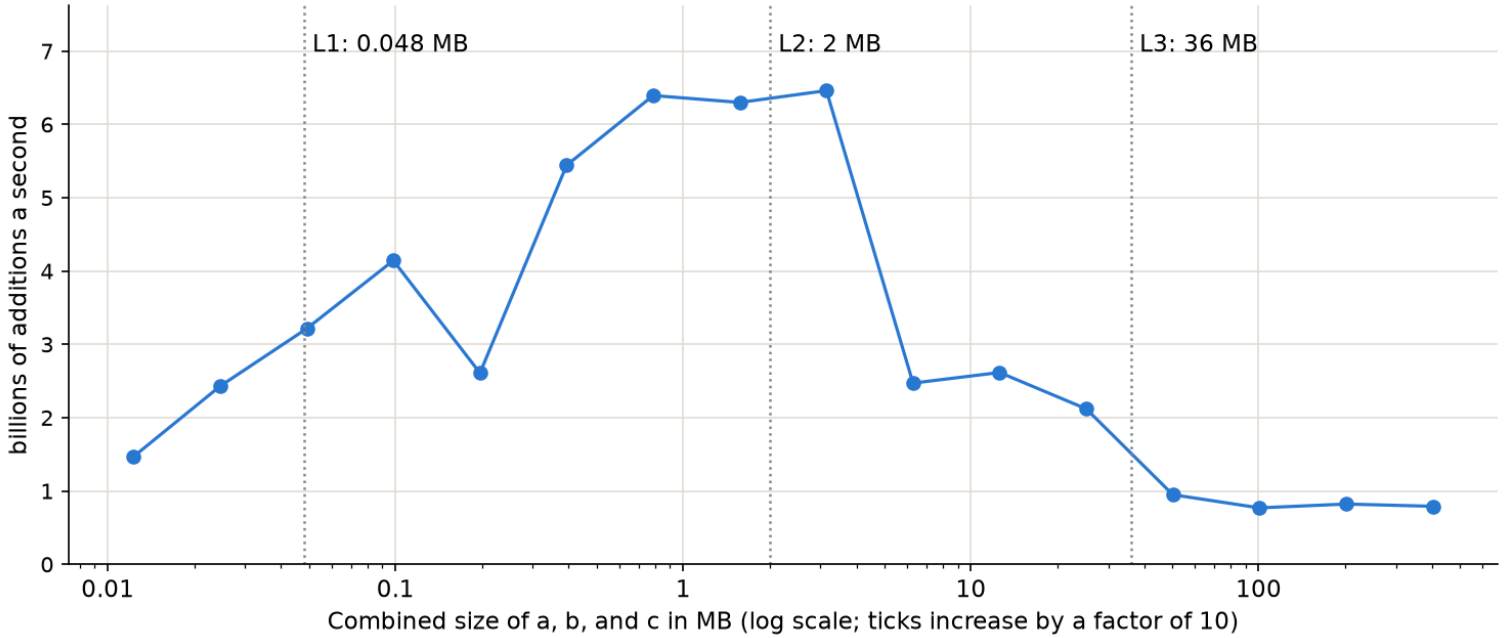
CS 301 · MEETING 12 · OPENING

The Memory of a Computer

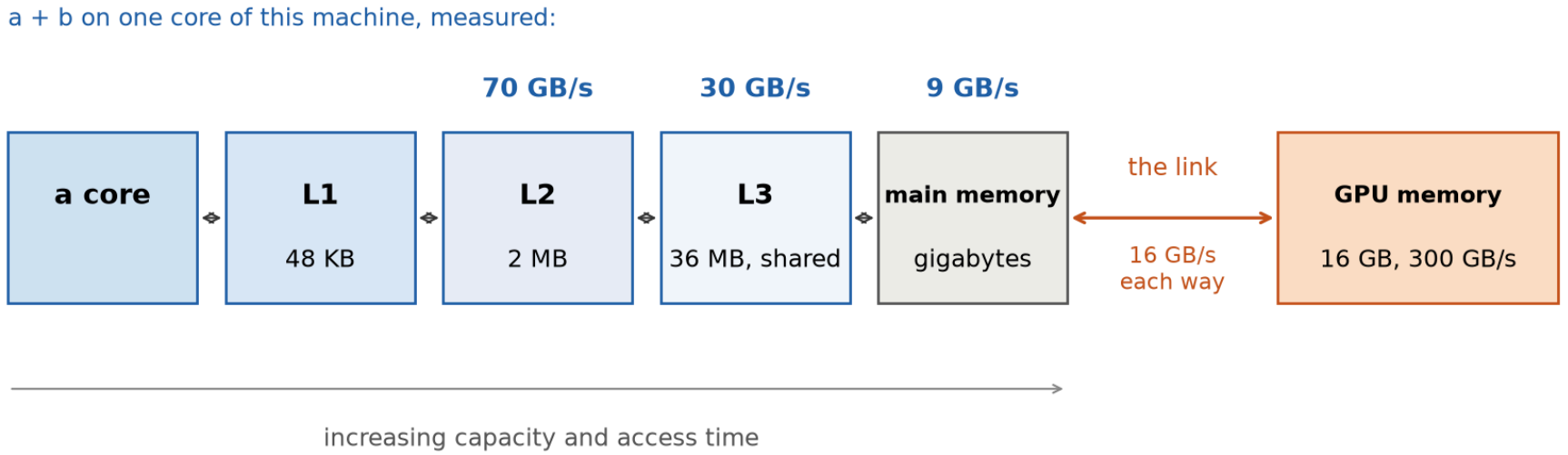
The same addition at different speeds: cache capacity, the memory hierarchy, and measured bandwidth.

The same additions, sixteen sizes



- Notebook Part 5, on this laptop: best of five runs
- Rates fall as the arrays outgrow the caches

The memory hierarchy



- Caches: lower capacity and faster access than main memory
- GPU memory is separate; the connecting link has lower bandwidth

Measured memory bandwidth

the three arrays	capacity	additions a second	bytes a second
in L2	2 MB, one core	5.87 billion	70.5 billion
in L3	36 MB, shared	2.47 billion	29.7 billion
in main memory	gigabytes	791 million	9.49 billion

- Useful traffic: **12 bytes/addition**, two float32 inputs and one output
- The rows compare L2 cache, L3 cache, and main memory
- a + b is **memory-bound**: memory bandwidth limits its speed